MILITARY SPECIFICATION

TRANSFORMER, POWER, ISOLATION

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 This specification covers an isolation power transformer, used in airfield lighting installations.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and publications, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-C-5136	Cable, Power, Electrical, Polychloroprene Sheathed, Buna Compound Insulated
MIL-P-7936	Parts and Equipment, Procedures for Packaging and Packing of
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Military

MIL-STD-129	Marking For Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property

FSC 5950

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Military (Continued)

MIL-STD-143	Specification and Standards, Order of Precedence for The Selection of
MS33586	Metals, Definition of Dissimilar

When requesting applicable documents refer to both title and number. Copies of this specification and applicable specifications and standards may be obtained upon application to the Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania, Code DCI.

2.2 Other publications - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Consolidated Classification Committee

Uniform Freight Classification Rules (Application for copies of the above publication should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station; Chicago 6, Illinois.)

3. REQUIREMENTS

3.1 Preproduction - The Isolation Power Transformer furnished under this specification shall be a product which has been tested and passed the preproduction tests specified herein.

3.2 Components - The transformer shall consist of primary and secondary coils wound upon a core and inclosed in a waterproof case with rubber connectors molded on the primary and secondary leads.

3.3 Selection of specifications and standards - Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with Standard MIL-STD-143 except as provided in 3.3.1 and 3.3.2.

3.3.1 Commercial parts - Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, such as screws, bolts, nuts, and cotter pins, having suitable properties may be used provided:

(a) They can be replaced by the standard parts without alteration.

(b) The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractor's drawings.

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3.3.2 Standard parts - With the exception in 3.3.1, standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.4 Materials -

3.4.1 Fungus-proof materials - Materials that are nutrients for fungi shall not be used where it is practical to avoid such materials. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed inclosure, fungicidal treatment will not be necessary.

3.4.2 Metals - Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions as may be encountered in storage or normal service.

3.4.2.1 Dissimilar metals - Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in Standard MS33586.

3.5 Design and construction - The transformer shall be so constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, and other conditions incident to shipping, storage, installation, and service. The exact shape and design of the transformer shall be optional, provided all requirements specified herein are met.

3.5.1 Rating - The transformer shall be designed to operate three 500w, 120v, Par 64 lamps at 180v, from a 2400v, 60-cycle line under an intermittent flashing condition. The lamps will be flashed on a duty cycle of 0.3 seconds "on" and 0.3 seconds "off" for periods not exceeding one minute in fifteen.

3.5.1.1 Windings - The windings shall be completely insulated from each other and the core. The transformer shall be designed to operate on load (16 ohms) for 1 hour or indefinitely on open-circuit conditions in the secondary with rated voltage and frequency in the primary. The transformer shall have sufficient capacity to allow for lamp aging and secondary contact losses.

3.5.2 Operating conditions - The transformer shall be constructed for continuous outdoor service at any ambient temperature from a minimum of -55°C to a maximum of +55°C, either on the surface of or buried directly in the ground or submerged in water.

3.6 Case - The transformer shall be inclosed in a permanently sealed case so as to produce a completely watertight assembly. The case shall be molded from rubber, or synthetic rubber, and may be molded directly on the transformer or premolded and compound filled. No portion of the case shall be less than 1/4 inch thick. Sharp corners and edges of the core and coil assembly shall be eliminated or adequate provisions made so that

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they will not cut the case if the transformer is dropped or handled roughly. The case shall be constructed so that moisture accidentally getting into the leads cannot be conducted through the leads into the transformer. Insofar as is practicable, internal airpockets shall be eliminated and the assembly shall be sufficiently rugged to withstand rough handling. The case shall be capable of withstanding exposure to sun, oil, gasoline, moisture, and acid and alkaline soils. The case shall form a permanent watertight junction with the sheath of the leads. The case shall be so designed that the transformer may be installed upright or lying on any side. The size of the transformer case shall be held to a minimum consistent with proper design. The junction of the case with the sheath of the leads shall not crack nor show evidence of damage when the transformer is handled by the leads nor when the leads are bent or twisted as they normally would be during installation.

3.7 Leads - The transformer shall be provided with a two-conductor secondary lead and two single-conductor primary leads. The leads shall be securely fastened to the transformer in such a manner that carrying by the leads shall not loosen the electrical connections nor damage the water seal.

3.7.1 Primary leads - The primary leads shall be single-conductor, 5,000v, No. 8AWG, stranded cable 30 ±3 inches long as measured from the junction with the transformer case without connectors.

3.7.2 Secondary lead - The secondary lead shall be 600v, two-conductor, No. 12AWG flexible cord, 54 ±3 inches long without connectors. Each conductor shall have not less than 37 strands. The lead shall be constructed of cable in accordance with Specification MIL-C-5136 or of similar high-quality commercial cable having synthetic-rubber insulation and a polychloroprene sheath.

3.8 Interchangeability - All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of Specification MIL-D-70327 shall govern changes in the manufacturer's part numbers.

3.9 Weight - The weight of the transformer shall be held to the minimum consistent with good design and shall not exceed 75 pounds.

3.10 Identification of product - The transformer shall be marked for identification in accordance with Standard MIL-STD-130, except the serial number shall not be included. The information shall be molded on the surface of the transformer or permanently marked on a nameplate attached to the transformer. If desired, part of the information may be molded on the transformer and the remainder included on a nameplate. The following special marking shall be added:

Multiple Type
Volts (primary and secondary)

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3.11 Workmanship - The transformer, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, welding and brazing, painting, riveting, machine-screw assemblies, and freedom of exposed parts from burrs and sharp edges.

3.11.1 Dimensions - Dimensions and tolerances, not specified, shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the transformer, they shall be held or limited accordingly.

3.11.2 Screw assemblies - Assembly screws and bolts shall be tight. The word tight means that the screw or bolt cannot be appreciably tightened further, without damage or injury to the screw, bolt, or threads.

3.11.3 Riveting - Riveting operation shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.11.4 Cleaning - The transformer shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign material, during and after final assembly. Burrs and sharp edges, as well as resin flash that may crumble, shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract of purchase order, the supplier is responsible for the performance of all inspection requirements as specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests - The inspection and testing of transformers shall be classified as follows:

(a) Preproduction tests (4.3)

(b) Acceptance tests (4.4)

4.3 Preproduction tests -

4.3.1 Tests - Preproduction tests shall consist of all the tests described under (4.6). Tests shall be made in the order specified.

4.3.2 Preproduction test sample - The preproduction test sample shall consist of one transformer.

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4.4 Acceptance tests - Acceptance tests shall consist of:

(a) Individual tests.

(b) Sampling plan and tests.

4.4.1 Individual tests - Each transformer shall be subjected to the following tests as described under (4.6):

(a) Examination of product

(b) Dielectric

(c) Ratio

4.4.2 Sampling plan and tests - From each lot of 100 or fraction thereof, a group of five transformers shall be selected at random from transformers that are representative of current production and that have passed the specified individual tests. This group of five transformers shall be subjected to the test specified in 4.6.6, except that the test procedures may be modified as specified in 4.6.2.

4.4.2.1 Individual tests may continue - For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or produced later shall not be made until it is determined that items meet all the requirements of the specification.

4.5 Test conditions -

4.5.1 Unless otherwise specified, the tests specified herein shall be conducted at room temperature (approximately 25°C).

4.5.2 The electrical tests specified herein shall be conducted using a suitable 60-cycle sine wave power source with primary voltage maintained at the specified values by means of an autotransformer.

4.6 Test methods -

4.6.1 Examination of product - The transformer shall be examined to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

4.6.1.1 The junction of the case and the leads shall be examined to determine that it has not been damaged and is otherwise satisfactory.

4.6.2 Dielectric - Each transformer shall be subjected to one complete cycle of the test specified in 4.6.6. The following modifications may be made in the test specified in 4.6.6:

(a) The heating cycle specified in 4.6.6.1 may be conducted by heating the transformers in an oven, or by short-circuiting either the primary or secondary coil

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and circulating slightly more than rated current through the other coil. The heating cycle, however, shall produce equivalent or greater temperatures in the coils, case, and leads of a transformer than the 5-hour full load (65 ohms) condition, and the transformers shall be heated for not less than 2 hours.

(b) Primaries of several transformers may be connected directly together and the transformers heated, soaked, and tested as a group, as for one transformer. The minimum, acceptance megohms insulation resistance for the combined primary coils of the group shall be not less than that specified for a single transformer. No transformer in the group shall be disconnected until after completion of the tests.

4.6.3 Ratio - With 2400v applied to the primary, the secondary voltage of each transformer at rated load shall be within the limits specified in table I.

TABLE I

Primary Voltage	Primary Power Factor (Min.) (Percent)	Efficiency (Minimum) Percent	Secondary Volts (Rms)	Load
2400	92	90	185 max.	65 ohms
2400	--	--	170 min.	16 ohms
2400	--	--	200 max.	Open circuited

4.6.4 Characteristics - The transformers shall be tested to determine compliance with the electrical characteristics specified in table I. All corrections necessary to compensate for meter-power consumption shall be applied. The open-circuit voltage of the transformer secondary shall be read by a voltmeter having a resistance of at least 100 ohms per volt.

4.6.5 Impact - Two of the transformers subjected to the test specified in 4.6.4 shall be dropped twice from a height of 4 feet upon a hard, rigid surface; once so they hit on a bottom corner or location where the most damage is likely to occur due to the core cutting into the case, and once so they hit on a side or location where the most damage is likely to occur to the windings: After being dropped, the two transformers shall be resubmitted to the test specified in 4.6.4. Any evidence of damage to the case or a change of more than 1 percent of either sample in the results obtained in the test specified in 4.6.4 shall cause the lot to be rejected.

4.6.6 Insulation resistance - The two transformers subjected to the test specified in 4.6.5 shall be subjected to a continuous, 20-cycle test.

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4.6.6.1 The transformers shall be operated, for a minimum of 5 hours in air, at room temperature, with 2400v applied to the primary and with 65 ohms on secondary.

4.6.6.2 The transformer shall be soaked in water, at room temperature, for not less than 12 hours, and the insulation resistance measurements repeated.

4.6.6.3 Measurements of dielectric and insulation resistance shall be made with direct current. The following voltage shall be applied for 1 minute between each coil and ground, with the other coil grounded and connectors submerged in water. The insulation resistance shall equal or exceed the values specified in table II. Zero and maximum readings of the test instrument shall be periodically checked by immersing the high voltage lead in water along side a transformer and by suspending it in air.

TABLE II

Coil	DC Test Voltage	Minimum Insulation Resistance in Megohms	
		Cold	Hot
Secondary	3,000	1,000	300
Primary	15,000	2,000	750

4.6.6.4 An insulation resistance less than the values specified in table II, a softening or swelling of the case, or any evidence of leakage or damage shall be cause for rejection of the lot.

4.6.7 Temperature rise - The temperature rise of the transformer not subjected to the impact test shall be determined by the resistance method and shall not exceed 55°C when the transformer is operated under any of the following conditions with 2400v applied to the primary.

(a) Rated load of 65 ohms.

(b) Open circuit.

4.6.8 The transformers shall be operated for a minimum of 1 hour in air, at room temperature, with 2400 volts applied to the primary and with three 500 watt 120 volt lamps connected to the secondary. Either the primary or the secondary circuit shall be interrupted using a cycle of 50% on and 50% off with a frequency of 90 - 110 cycles per minute. Upon completion of the tests, the insulation resistance shall be determined by the method specified in paragraph 4.6.6.3. The cold insulation resistance shall not be less than those specified in table II.

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4.6.9 Internal inspection - One transformer shall be cut open and examined internally to determine compliance with all applicable requirements pertaining to materials, and method of construction.

4.6.10 Inspection of the preservation, packaging, packing, and marking for shipment and storage - Sample items or packs and the inspection of the preservation, packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of Section 5, or the documents specified therein.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging - Preservation and packaging shall be Level A or C as specified (see 6.2).

5.1.1 Level A - The transformer shall be preserved in accordance with Specification MIL-P-116, method IC. Unit quantities shall be one each.

5.1.2 Level C - The transformer shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.2 Packing - Packing shall conform to Specification MIL-P-7936, Level A, B, or C, as specified (see 6.2).

5.3 Marking - Interior and exterior containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended use - The transformer covered by this specification is intended for use in a wheels-up warning system in both permanent and advance base runway lighting installations.

6.2 Ordering data - Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Selection of applicable levels of packaging and packing (see Section 5).

(c) Quantity of transformers required.

6.3 Drawing requirements - The attention of contractors is invited to the fact that Government contracts for equipment, whether procured directly by the Government or through another contractor, require that all drawings submitted must be in accordance with the requirements of Specification MIL-D-70327, or other applicable drawing specifications.

6.4 Provisions for preproduction tests - Quantity production on a particular contract should be withheld until the preproduction sample has been pronounced satisfactory by the procuring activity. When a contractor

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is in continuous production of these transformers from contract to contract, submission of additional preproduction samples on the subsequent contracts may be waived at the discretion of the procuring activity. Approval of preproduction samples or the waiving of preproduction tests does not preclude the requirements of acceptance testing.

6.4.1 It is understood that the transformers supplied under contract will be identical in every respect to the corresponding approved preproduction sample and to the drawings except for changes which have been previously approved by the Government.

6.4.2 Evidence of unauthorized change from the approved preproduction sample or drawings will constitute cause for rejection.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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